

Logged in as **doctor1**

Role: **Doctor**

Log out



Obesity Level Predictor

The predictions are based on specific training data and can be incorrect.The app is aimed to help clinicians and assist them, not replace them.Regardless of the outcome, it is advised to consult to a doctor.

Patient Profile

Demographics & Habits

Gender

Male

Family history of overweight

no

Frequent high-caloric food (FAVC)

yes

Eating between meals (CAEC)

no

Smoker

yes

Monitors calorie intake (SCC)

yes

Alcohol consumption (CALC)

Frequently

Transportation used (MTRANS)

Automobile

Measurements & Frequencies

Age (years)

25.00

Height (m)

1.60

Weight (kg)

77.50

Vegetable consumption frequency (FCVC)

2.00

Number of main meals (NCP)

3.00

Daily water intake (CH2O)

1.00

Physical activity frequency (FAF)

2.00

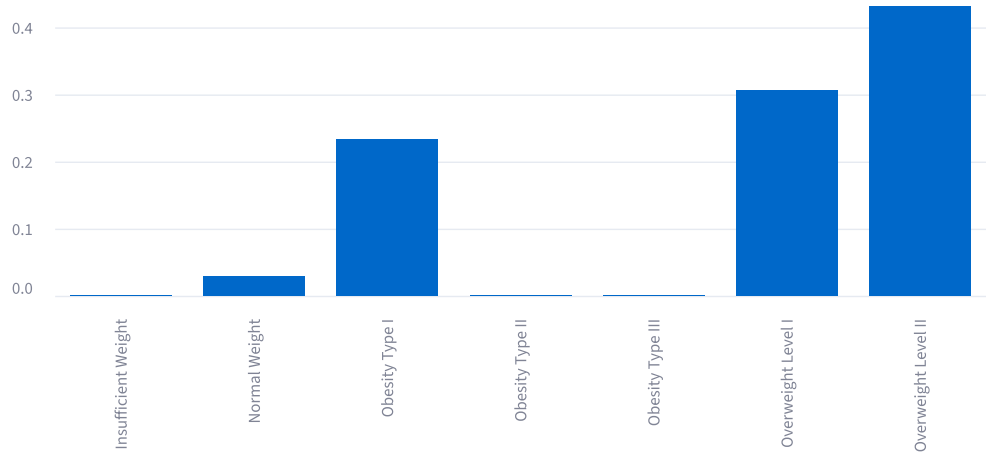
Time using technology devices (TUE)

1.00

Predict

Prediction

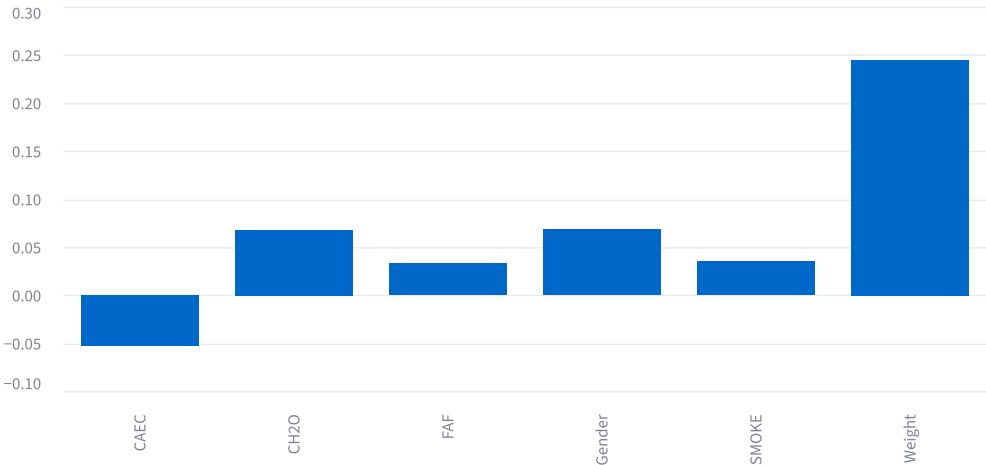
Predicted class: **Overweight Level II**



Class	Probability
Overweight Level II	43.08%
Overweight Level I	30.64%
Obesity Type I	23.29%
Normal Weight	2.94%
Obesity Type II	0.04%
Insufficient Weight	0.00%
Obesity Type III	0.00%

Why this prediction?

Top features pushing the model toward **Overweight Level II**, estimated with SHAP against a background sample from the train+val pool.



Feature	Current Value	Contribution	Direction
Weight	77.500000	+0.245	toward Overweight Level II
Gender	Male	+0.068	toward Overweight Level II
CH2O	1.000000	+0.068	toward Overweight Level II
CAEC	no	-0.052	away from Overweight Level II
SMOKE	yes	+0.035	toward Overweight Level II
FAF	2.000000	+0.033	toward Overweight Level II

▼ Show submitted patient profile

Gender	family_history_with_overweight	FAVC	CAEC	SMOKE	SCC	CALC	MTRANS	Age
Male	no	yes	no	yes	yes	Freque	Automobile	25

Similar Patient Profiles (top 5)

Nearest neighbors in the model's learned latent space, from the train+val pool.

latent_distance	Gender	family_history_with_overweight	FAVC	CAEC	SMOKE	SCC	CALC
2.621	Male	no	yes	Sometimes	no	no	Sometimes
2.674	Male	no	yes	Sometimes	no	no	Frequently
2.739	Male	no	yes	Sometimes	no	yes	no
2.884	Female	yes	yes	Sometimes	no	no	no
2.899	Female	yes	yes	Sometimes	yes	no	Sometimes

Healthier Prototype

Predicted class **Overweight Level II** → one step healthier: **Overweight Level I**

Candidates with Gender=Male, Age in [23.0, 27.0], class=Overweight_Level_I: 30

Top 7 nearest 'Overweight Level I' neighbors (same gender, age ±2):

feature_distance	Gender	family_history_with_overweight	FAVC	CAEC	SMOKE	SCC	CALC
4.072	Male	yes	yes	Sometimes	no	no	Sometime
4.210	Male	no	no	Sometimes	no	no	no
4.243	Male	yes	yes	Sometimes	no	no	Sometime
4.270	Male	no	yes	Sometimes	no	no	no
4.300	Male	no	yes	Sometimes	no	no	no
4.333	Male	yes	yes	Sometimes	no	no	Sometime
4.441	Male	no	yes	Sometimes	no	no	no

Recommended Changes

Excluded non-actionable features: Age, Height, Gender, family_history_with_overweight. Ranked by how much each feature would need to shift to look like a healthier profile.

Obesity Level Predictor

feature	current_value	prototype_value	shift_magnitude
SMOKE	yes	no (100% of neighbors)	1.414
SCC	yes	no (100% of neighbors)	1.414
CAEC	no	Sometimes (100% of neighbors)	1.414
CH2O	1.000000	1.800000	1.313
CALC	Frequently	no (57% of neighbors)	1.229
TUE	1.000000	0.480000	0.867
FCVC	2.000000	2.340000	0.636
FAF	2.000000	1.530000	0.555